

1. Question:

The Prophet (PBUH) said: "If one does not abandon falsehood, God has no need that one should abandon food and drink" (Bukhari). How does the self-discipline and Taqwa gained through fasting reflect in the ethical conduct of a software engineer, particularly in avoiding data manipulation, online falsehood, and digital misconduct? Support your answer with Quranic references.

Answer:

Fasting (Sawm) is not only about avoiding food and drink, but also about developing self-discipline and Taqwa (God-consciousness). The Hadith shows that if a person continues lying or doing wrong actions, then fasting loses its real purpose.

During Ramadan, a Muslim learns to control desires, speak truth, and avoid harmful behavior. This inner discipline directly influences the ethical conduct of a software engineer. In the digital world, engineers deal with sensitive data, systems, and information. Taqwa helps them remember that Allah is always watching, even when no one else is.

For example, a software engineer with Taqwa will avoid **data manipulation**, such as changing user data for personal benefit or company profit. This is because Islam clearly prohibits dishonesty. Allah says:

👉 “Do not consume one another’s wealth unjustly” (Qur’an 2:188).

Similarly, fasting teaches a person to avoid **false speech and online lies**. This applies to spreading misinformation, fake content, or misleading users through algorithms. The Qur’an warns:

👉 “Do not spread corruption on the earth” (Qur’an 2:60).

A Muslim engineer will also avoid **digital misconduct**, such as hacking, privacy violations, or unethical AI use. In Islam, user data is considered an Amanah (trust), so it must be protected.

Fasting builds patience and accountability. When a person can control hunger and thirst for Allah, they can also control unethical behavior in professional life. This leads to honesty, transparency, and responsibility in software development.

In conclusion, fasting develops Taqwa, which guides a software engineer to act

ethically, avoid falsehood, and protect users. Thus, fasting transforms both personal character and professional conduct.

2. Question:

Define fasting (Sawm) and explain its philosophical basis. How does the concept of fasting as a "month-long training period" for self-control relate to the professional discipline required of a Muslim software engineer?

Answer:

Fasting (Sawm) is defined as abstaining from eating, drinking, and other invalidating actions from dawn to sunset with the intention of worshipping Allah.

The philosophical basis of fasting is self-control and spiritual purification. Islam teaches that fasting helps a person control their desires, especially hunger and physical needs. By controlling these strong desires, a person becomes capable of controlling other aspects of life, such as speech, behavior, and thoughts.

Fasting is often described as a **"month-long training period"**. During Ramadan, Muslims practice discipline daily. They wake up early, avoid prohibited actions, control anger, and increase good deeds. This repeated practice builds habits of patience, focus, and responsibility.

This concept is directly related to the professional discipline required of a Muslim software engineer. In software engineering, discipline is very important. Engineers must write correct code, follow ethical rules, and avoid shortcuts that can harm users or systems.

For example, just as a fasting person avoids food even when no one is watching, a disciplined engineer avoids unethical actions like copying code illegally, using pirated software, or inserting hidden harmful functions.

Fasting also teaches **time management**, as Muslims follow strict schedules for Suhoor, Iftar, and prayers. This helps engineers manage deadlines, projects, and responsibilities effectively.

Moreover, fasting develops **patience and emotional control**. In professional life, engineers face stress, bugs, and pressure. The patience learned in Ramadan

helps them stay calm and solve problems logically instead of reacting emotionally.

Finally, fasting strengthens the awareness of Allah (Taqwa). This ensures that a Muslim engineer remains honest and accountable, even when working alone.

In conclusion, fasting is not just a ritual but a training system. It builds self-control, discipline, and responsibility, which are essential qualities for a professional and ethical software engineer.

3. Question:

The Prophet (PBUH) said: "Whoever can guarantee what is between his two jaw-bones and what is between his two legs, I guarantee Paradise for him" (Bukhari). Explain the physical and spiritual benefits of fasting, including the concept of Autophagy, and discuss how bodily and mental discipline gained through Ramadan shapes a Muslim engineer's professional character.

Answer:

The Hadith emphasizes controlling two important things: the tongue (speech) and desires. Fasting helps achieve this control by developing both physical and spiritual discipline.

Physically, fasting has many benefits. One important concept is **Autophagy**, which is a process where the body cleans and renews damaged cells. Scientific research shows that fasting activates autophagy, which improves health and slows aging.

Fasting also improves digestion, controls body weight, and reduces harmful habits like overeating. This leads to a healthier body, which is important for maintaining productivity and focus in professional life.

Spiritually, fasting strengthens Taqwa and self-discipline. A fasting person avoids lying, gossiping, and harmful actions. This develops control over speech, as mentioned in the Hadith.

These physical and spiritual benefits shape a Muslim software engineer's professional character in many ways. First, controlling the tongue helps the

engineer avoid harmful communication, such as spreading false information or insulting others online.

Second, controlling desires helps avoid unethical actions like greed, corruption, or misuse of technology. For example, an engineer will not create harmful software, even if it offers high profit.

Third, fasting builds **mental strength and focus**. Engineers often need deep concentration for coding and problem-solving. The discipline gained during Ramadan improves attention and reduces distractions.

Finally, fasting teaches accountability and sincerity. A Muslim engineer understands that all actions are seen by Allah, so they act with honesty and responsibility in their work.

In conclusion, fasting improves both body and soul. Through physical benefits like autophagy and spiritual growth like self-control, it shapes a disciplined, ethical, and responsible software engineer.

4. Question:

Hajj unites Muslims across all nationalities, colors, and social classes, reflecting "the noblest of you is the best in conduct" (49:13). Drawing from the spiritual dimensions of Hajj, discuss how a Muslim software engineer should uphold fairness, inclusivity, and brotherhood when building global digital platforms and open-source systems.

Answer:

Hajj is one of the most powerful examples of unity in Islam. Millions of Muslims from different countries, races, and social classes gather together and perform the same rituals. This shows that all humans are equal, and the only superiority is based on good character and conduct, as stated in the Qur'an:

👉 **"The noblest of you in the sight of Allah is the most righteous" (49:13).**

The spiritual dimension of Hajj teaches equality, humility, and brotherhood. Everyone wears simple Ihram, removing all differences of wealth and status. This lesson can be applied by a Muslim software engineer when building global digital platforms.

First, the principle of **fairness** means that software systems should treat all users equally. Engineers should design algorithms that do not discriminate based on race, religion, gender, or nationality. For example, AI systems must avoid bias and ensure equal opportunities for all users.

Second, **inclusivity** is very important. Just like Hajj welcomes all Muslims, digital platforms should be accessible to people from different backgrounds. This includes supporting multiple languages, designing for people with disabilities, and making technology affordable.

Third, the concept of **brotherhood** encourages collaboration and mutual respect. In open-source systems, developers from around the world work together. A Muslim engineer should contribute honestly, share knowledge, and help others instead of seeking only personal benefit.

Hajj also teaches patience, cooperation, and respect for others. These qualities are essential in teamwork and global development environments. Engineers should avoid harmful content, hate speech, and division in online platforms.

In conclusion, the spiritual lessons of Hajj guide Muslim software engineers to build fair, inclusive, and ethical systems that promote unity and benefit all humanity.

5. Question:

Explain the historical background of Hajj and its significance as the 5th pillar of Islam. Discuss the Pillars (Arkan) and Wajib (Obligatory) acts of Hajj, and explain how the spirit of complete submission demonstrated in Hajj applies to a Muslim engineer's obedience to Islamic ethics in their profession.

Answer:

Hajj has a deep historical background connected to Prophet Ibrahim (AS). Allah commanded him to leave his wife Hajar and son Ismail (AS) in the desert of Makkah. Later, Prophet Ibrahim and Ismail (AS) built the Ka'bah as a place of worship.

Hajj became an obligation in Islam and is considered the **5th pillar of Islam**. It is required once in a lifetime for those who are physically and financially able.

The main **Pillars (Arkan) of Hajj** are:

1. Ihram (intention and state of purity)
2. Standing at Arafat
3. Tawaf al-Ifadah
4. Sa'ee (walking between Safa and Marwa)

The **Wajib (Obligatory) acts** include:

- Ihram at Miqat
- Staying at Arafat until sunset
- Spending night in Muzdalifah
- Stoning at Jamarat
- Cutting or shaving hair
- Staying in Mina
- Tawaf al-Wada

Hajj represents complete submission to Allah. Many rituals may not seem logical at first, but Muslims perform them because Allah commanded them. This shows total obedience and trust.

This spirit of submission is very important for a Muslim software engineer. In professional life, engineers may face situations where unethical work is profitable, such as developing harmful applications or engaging in dishonest practices.

A Muslim engineer should follow Islamic ethics just like a pilgrim follows Hajj rituals. Even if something is allowed legally but prohibited in Islam (such as Riba-based systems or harmful software), the engineer must avoid it.

Hajj teaches obedience, discipline, and sincerity. These values help engineers remain honest, follow ethical guidelines, and prioritize Halal over personal gain.

In conclusion, Hajj is not only a ritual but a lesson of submission, which guides Muslim engineers to remain ethical and obedient to Islamic principles in their professional life.

6. Question:

Islam positions humans as Khalifah (stewards) of the earth, commanded "Do not commit abuse on the earth" (2:60). As a CSE student whose work contributes to energy-intensive data centers and e-waste, discuss your environmental responsibilities using the Islamic principles of Khalifah, Mizan (balance), and resource conservation, with references from the Quran and Sunnah.

Answer:

In Islam, humans are considered **Khalifah (stewards)** of the earth. This means they are responsible for protecting and managing the environment properly. Allah commands:

👉 “Do not commit abuse on the earth” (Qur’an 2:60).

This responsibility is very important for CSE students and software engineers, as modern technology contributes to environmental issues like energy consumption and electronic waste.

The concept of **Khalifah** means that we must use resources responsibly and avoid harming the environment. Technology should be used for benefit, not destruction.

Another important principle is **Mizan (balance)**. Islam teaches that everything in the universe is created in balance. Humans should not disturb this balance by overusing resources or causing pollution.

The Qur’an also says:

👉 “Eat and drink but do not waste, for Allah does not like the wasters” (7:31).

This applies to energy use in data centers and excessive digital consumption. Engineers should design energy-efficient systems and reduce unnecessary processing and storage.

The principle of **resource conservation** is also emphasized in Islam. The Prophet (PBUH) taught not to waste water even when it is abundant.

As a CSE student, practical responsibilities include:

- Developing energy-efficient algorithms
- Promoting green computing
- Reducing e-waste through recycling and sustainable design
- Avoiding unnecessary data storage and processing

Islam also encourages environmental care through good actions like planting trees, which are considered ongoing charity (Sadaqah).

In conclusion, the principles of Khalifah, Mizan, and conservation guide Muslim engineers to protect the environment. By applying these values, they can contribute to sustainable technology and fulfill their responsibility as stewards of the earth.

7. Question:

The Prophet (PBUH) said: "Whoever plants a tree and diligently looks after it until it matures and bears fruit is rewarded" (Musnad). Discuss the Islamic approach to environmental sustainability and resource conservation. How can Muslim software engineers contribute to green computing and sustainable technology development in light of these teachings?

Answer:

Islam strongly emphasizes environmental sustainability and responsible use of natural resources. The Hadith about planting trees shows that caring for the environment is considered a continuous charity (Sadaqah Jariyah).

The Islamic approach to sustainability is based on balance, responsibility, and long-term benefit. Humans are not owners of the earth but trustees. Therefore, they must protect nature and avoid waste. The Qur'an warns against misuse of resources:

👉 **"Do not commit corruption on the earth" (2:60).**

Islam also teaches moderation. Overconsumption and waste are discouraged:

👉 **"Allah does not like the wasters" (7:31).**

The Prophet (PBUH) also showed practical examples of conservation, such as avoiding waste of water even when it is abundant.

In modern times, software engineers play an important role in environmental sustainability. Technology consumes large amounts of energy and produces electronic waste. Therefore, Muslim software engineers should follow Islamic teachings and contribute to **green computing**.

First, they can develop **energy-efficient software** by optimizing algorithms and reducing unnecessary processing. This lowers the energy consumption of data centers.

Second, they should promote **sustainable system design**, such as reducing storage waste, using cloud resources efficiently, and encouraging recycling of hardware.

Third, engineers can support **environment-friendly technologies**, such as smart energy systems and applications that reduce carbon emissions.

Finally, they should raise awareness about sustainability in the tech community, just as Islam encourages spreading good practices.

In conclusion, Islam promotes environmental protection through responsibility, balance, and conservation. Muslim software engineers can apply these principles by developing sustainable technologies and supporting green computing for the benefit of society and the environment.

8. Question:

Islam classifies punishment into Hadd, Qisas, and Ta'zir, holding every intellectually capable person accountable for distinguishing right from wrong. Using the objectives and distinguishing features of the Islamic Penal System, analyze which punishment category best applies to cybercrimes such as hacking, ransomware, and identity theft, and justify your answer with references.

Answer:

Islamic law classifies punishment into three main categories: **Hadd, Qisas, and Ta'zir**. Hadd refers to fixed punishments defined by Allah, Qisas refers to equal retaliation, and Ta'zir refers to discretionary punishments decided by authorities.

Cybercrimes such as hacking, ransomware, and identity theft are modern issues that are not directly mentioned in classical Islamic texts. Therefore, they do not fall under Hadd or Qisas. Instead, they are best classified under **Ta'zir punishment**.

Ta'zir is flexible and is applied to crimes that harm individuals or society but do not have fixed punishments. It allows judges or authorities to decide appropriate penalties based on the situation.

The objectives of the Islamic Penal System include protecting five essential elements: **life, religion, intellect, lineage, and wealth.**

Cybercrimes clearly violate these objectives, especially the protection of wealth and privacy. For example:

- **Hacking** can damage systems and steal information
- **Ransomware** causes financial loss and fear
- **Identity theft** violates personal rights and security

Islam aims to maintain justice and social order. The Qur'an states:

👉 “We sent down the Book and the balance so that people may establish justice” (57:25).

The Islamic Penal System also focuses on both **deterrence and reform.**

Punishments are designed to prevent crime and guide people toward correct behavior.

Since cybercrimes are complex and constantly evolving, Ta'zir is the most suitable category because it allows adaptation to new situations. Authorities can decide punishments such as fines, imprisonment, or restrictions depending on the severity of the crime.

In conclusion, cybercrimes fall under Ta'zir because they are modern offenses not covered by fixed punishments. This flexible system ensures justice, protects society, and aligns with the objectives of Islamic law.

9. Question:

Explain the concept of Al-Thawab Wa Al-Iqab (Reward and Punishment) in Islam. How does the Islamic belief in divine accountability motivate a Muslim software engineer to maintain ethical standards in their professional conduct, even when no one is watching? Answer:

Al-Thawab Wa Al-Iqab refers to the concept of **reward and punishment** in Islam. It means that every good action is rewarded by Allah, and every bad action is punished. This system is based on human free will and accountability.

Islam teaches that life is a test, and humans are responsible for their choices. Allah rewards good deeds and punishes wrong actions. The Qur'an emphasizes justice and accountability, reminding believers that nothing is hidden from Allah.

This belief has a strong impact on a Muslim software engineer's professional behavior. In many cases, engineers work alone or have access to sensitive systems without supervision. However, belief in divine accountability ensures that they remain honest and responsible.

For example, a Muslim engineer will avoid unethical actions such as:

- Manipulating data
- Using pirated software
- Violating user privacy
- Creating harmful applications

Even if these actions are not detected by others, the engineer knows that Allah is always aware. This creates a strong internal control system, which is more effective than external supervision.

The Islamic system also combines **hope and fear**. A person hopes for reward (Thawab) and fears punishment (Iqab). This balance encourages continuous ethical behavior.

Furthermore, Islam encourages repentance. If a person makes a mistake, they can correct themselves and return to the right path. This promotes personal improvement and moral development.

In professional life, this belief leads to qualities such as honesty, trustworthiness, and accountability. A Muslim engineer treats their work as a responsibility (Amanah) and ensures that their actions benefit society.

In conclusion, the concept of reward and punishment creates a strong moral foundation. It motivates Muslim software engineers to maintain high ethical standards, even when no one is watching, because they are always accountable to Allah.

10. Question:

As a Muslim software engineer, you are offered a lucrative contract to build a peer-to-peer lending platform that charges compound interest (Riba). Using the Islamic principles of Halal and Haram — including the rule "What leads to Haram is itself Haram" — justify your decision to accept or reject the project. Additionally, propose a Shariah-compliant alternative architecture for the platform.

Answer:

As a Muslim software engineer, I would **reject** the contract to build a peer-to-peer lending platform that charges compound interest, because it is based on **Riba**, which is Haram in Islam. The slides clearly mention that developing **interest-based lending platforms** falls under Haram in software development.

In Islam, **Halal** means what is lawful and permitted, while **Haram** means what is forbidden by Allah and His Messenger. The Qur'an allows what is lawful and good, and Islamic teachings clearly separate Halal from Haram. The course also teaches an important rule of Islamic jurisprudence: "**What leads to Haram is itself Haram.**" This means that even if I am only writing the code, my work would directly support a Haram financial system. Therefore, the income from such a project would also be doubtful or unlawful.

The slides also emphasize that a Muslim engineer must think about the **purpose, impact, justice, and accountability before Allah** of the technology they build. A Riba-based platform promotes financial injustice and exploitation, especially when compound interest increases the burden on borrowers. So, accepting such a project would go against Islamic ethics and professional responsibility.

A better option is to propose a **Shariah-compliant alternative architecture**. Instead of interest-based lending, the platform could use:

- **Qard Hasan:** interest-free loans for people in need
- **Musharakah:** profit-and-loss sharing partnership
- **Mudarabah:** one party provides capital and the other provides effort, and profits are shared fairly
- **Murabaha-style financing:** the platform helps purchase a product and earns a transparent service fee instead of interest

Technically, this alternative platform could include:

- investor and borrower profiles
- contract templates based on Islamic finance models
- transparent profit-sharing calculation
- no compound interest engine
- clear audit logs and disclosures
- Shariah review and compliance layer

In conclusion, I would reject the original project because it supports Riba, which is Haram. Instead, I would recommend a Shariah-compliant fintech platform based on fairness, transparency, and lawful profit.

11. Question:

Explain the concept of Amanah (Trust) in the context of data privacy and cybersecurity. A company you work for suffers a data breach exposing thousands of users' personal and financial data. From an Islamic ethical standpoint, what are your responsibilities as the lead developer? Reference relevant Quranic principles in your answer.

Answer:

In Islam, **Amanah** means trust, responsibility, and something placed in a person's care that must be protected. In the context of data privacy and cybersecurity, user data is an Amanah. The slides clearly state that personal data must be treated as a trust and must be protected properly.

The Qur'an says:

👉 **"Indeed, Allah commands you to render trusts to whom they are due"**
(Qur'an 4:58).

This verse shows that protecting people's information is not only a technical duty but also a moral and religious responsibility.

If a data breach exposes thousands of users' personal and financial data, my responsibilities as the lead developer are serious. First, I must act with **honesty and transparency**. I should not hide the breach or give false information. The slides mention that breaches must be reported honestly and promptly.

Second, I must take immediate technical action to reduce further harm. This includes:

- identifying the source of the breach
- patching the vulnerability
- securing databases and servers
- resetting access credentials
- isolating affected systems if needed

Third, I must inform management and affected users clearly, because hiding the truth would violate Amanah and could increase damage. Islam strongly opposes cheating, deception, and harm. The course also highlights the principle: **“No harm and no reciprocating harm.”**

Fourth, I should review the system and improve future protection by applying strong encryption, minimal data collection, better access control, and regular security audits. The slides clearly say that engineers must collect only necessary data, protect it strongly, and never misuse or sell it.

Finally, I must take responsibility, learn from the incident, and work to restore trust. In Islam, leadership means accountability, not excuse-making.

In conclusion, Amanah in data privacy means that user data must be protected with honesty, care, and responsibility. After a breach, a Muslim lead developer must respond truthfully, reduce harm, fix the system, inform users, and strengthen security in light of Qur’anic ethics.

12. Question:

Define Halal and Haram with references from the Quran and Hadith. Explain the five categories of Islamic rulings (Fard, Mustahabb, Mubah, Makruh, Haram) and apply them to classify the following software development activities: developing an Islamic prayer app, building a gambling platform, creating a cybersecurity tool, and designing a social media algorithm.

Answer:

Halal means what is lawful, permissible, and allowed in Islam. The slides define Halal as anything allowed according to Islamic law. The Qur’an says:

👉 **“O mankind! Eat of that which is lawful and good on the earth”** (Surah Al-Baqarah, 2:168).

Haram means what is forbidden or prohibited in Islam. The slides also quote the Hadith:

👉 **“The halal is clear and the haram is clear, and between them are matters unclear...”** (Sahih al-Bukhari and Muslim).

The five categories of Islamic rulings are:

1. **Fard/Wajib** – obligatory, must be done
2. **Mustahabb/Sunnah** – recommended, rewarded if done
3. **Mubah** – permissible or neutral
4. **Makruh** – disliked, better to avoid
5. **Haram** – forbidden, must be avoided

Now these can be applied to software development activities:

1. Developing an Islamic prayer app

This can be classified as **Mustahabb**, and in some cases very close to a highly beneficial act, because it helps Muslims pray, learn Islam, and protect religion. The slides list Islamic apps such as Qur'an, prayer time, and Qibla apps under Halal software development.

2. Building a gambling platform

This is clearly **Haram**. The slides explicitly list gambling and betting platforms under Haram software development.

3. Creating a cybersecurity tool

This is usually **Mubah** or **Mustahabb**, depending on the purpose. If it is used to protect systems, prevent attacks, and defend people's data, it is beneficial and may be Mustahabb. But if it is designed for malicious hacking, then it becomes Haram. The slides distinguish between ethical cybersecurity and harmful hacking.

4. Designing a social media algorithm

This is generally a **gray area**, so it may be **Mubah** or **Makruh**, depending on how it is used. If it promotes beneficial content and honest communication, it can be permissible. But if it spreads addiction, division, lies, or harmful content, it may become Makruh or even Haram. The slides mention social media and algorithms as areas needing careful ethical analysis.

In conclusion, Islam provides clear guidance through Halal and Haram, while the five rulings help classify actions more precisely. A Muslim software engineer must judge technology by its purpose, effect, and obedience to Allah.

13. Question:

Discuss the Islamic ruling on software piracy and intellectual property. Using the Hadith "Whoever cheats us is not one of us" and the Quranic principle "Do not consume one another's wealth unjustly" (2:188), analyze whether using cracked software is permissible and suggest Halal alternatives available to CSE students.

Answer:

In Islam, software piracy is not permissible because it violates the rights of others and involves dishonesty. The slides clearly state that **intellectual property is a form of wealth protected in Islam**. Therefore, using cracked or pirated software is a kind of unlawful taking of someone else's property.

The Qur'anic principle says:

👉 **"Do not consume one another's wealth unjustly" (Qur'an 2:188).**

This verse applies not only to physical property but also to digital property, such as software, licenses, and paid tools. A company or developer spends time, money, and effort to create software, so using it without permission is unjust.

The Hadith also says:

👉 **"Whoever cheats us is not one of us" (Sahih Muslim).**

Using cracked software involves deception because a person is getting access to a paid product without paying for it. This is why the slides directly classify **using cracked or pirated software** and **sharing paid software without permission** as Haram.

For CSE students, the argument is sometimes made that software is expensive and needed for education. However, financial difficulty does not make piracy lawful when Halal alternatives exist. Islam encourages lawful means even when the easier unlawful option is available.

The slides suggest several **Halal alternatives** for students:

- **Open-source software** such as tools under MIT, GPL, or Apache licenses

- **Legitimate student discounts**
- **Free trial versions used properly**
- **Free alternatives**, such as GIMP instead of Photoshop

Students can also use educational licenses, web-based free platforms, and community editions of professional tools. These options allow learning without violating Islamic ethics.

In conclusion, using cracked software is not permissible in Islam because it is a form of cheating and unjust consumption of wealth. A Muslim CSE student should choose Halal alternatives such as open-source tools, student licenses, and properly used free versions.

14. Question:

Discuss the ethical considerations of Artificial Intelligence and Machine Learning from an Islamic perspective. Using the principles of Halal and Haram, differentiate between beneficial AI applications and harmful ones, and explain how the concept of accountability before Allah applies to AI developers.

Answer:

From an Islamic perspective, Artificial Intelligence (AI) and Machine Learning (ML) are tools. Like other technologies, they are not automatically Halal or Haram by themselves. Their ruling depends on **purpose, use, benefit, and harm**. The slides clearly say that technology is neutral, and its permissibility depends on how it is used.

Beneficial AI applications are those that serve humanity and promote justice, welfare, and protection. The slides list examples such as:

- **Healthcare diagnosis**
- **Disaster prediction**
- **Accessibility tools** for people with disabilities
- systems that improve useful services and help society

These applications can be considered Halal because they support good objectives and reduce harm. They may even serve the Islamic goals of protecting life, intellect, and public welfare.

On the other hand, harmful AI applications are Haram because they cause injustice, deception, or damage. The slides mention examples such as:

- **Autonomous weapons**
- **Mass surveillance for oppression**
- **Deepfakes used for deception**
- **Biased AI** that discriminates against people
- AI that violates privacy or human dignity

Islam also requires **transparency, fairness, accountability, and privacy protection** in technology. Users should know when they are interacting with AI, and developers must avoid systems that spread misinformation or unfair treatment.

A very important principle here is accountability before Allah. The slides remind engineers to ask whether they can stand before Allah for what they create. This means AI developers are morally responsible not only for writing code, but also for the social effects of their systems.

So, a Muslim AI developer must ask:

- Does this application benefit people or harm them?
- Does it protect privacy and fairness?
- Can it be misused for oppression or deception?

In conclusion, AI is Halal when it is used for beneficial, honest, and fair purposes, and Haram when it causes harm, injustice, or deception. Muslim AI developers must remain conscious that they are accountable to Allah for both the design and consequences of their systems.

15. Question:

Using the framework of Maqasid ash-Shariah (the five objectives: **Din**, **Nafs**, **Aql**, **Nasl**, and **Mal**), evaluate the ethical permissibility of developing an AI-powered facial recognition system intended for use by a government. Identify which Maqasid it serves and which it may violate, and state the conditions under which it would be considered Halal.

Answer:

Maqasid ash-Shariah refers to the higher objectives of Islamic law. The five main objectives are the protection of **Din (religion)**, **Nafs (life)**, **Aql (intellect)**, **Nasl (lineage)**, and **Mal (wealth)**. A facial recognition system should be evaluated by asking whether it serves or harms these objectives.

The slides provide a direct example of **AI facial recognition** and explain that it may be permissible if used ethically with strict privacy protections.

First, such a system may serve **Nafs (protection of life)** if it is used for security, finding criminals, preventing violence, or protecting people from danger.

Second, it may serve **Mal (protection of wealth)** if it helps prevent fraud, identity theft, or financial crime.

However, it can also violate several Maqasid. It may harm **Aql** and human dignity if people live under constant fear and pressure. It may threaten **Nasl** and family privacy if personal lives are monitored without justification. It may also indirectly affect **Din** if the system is used to oppress religious groups or limit freedom of lawful religious practice. Most importantly, misuse of facial recognition can violate privacy and lead to injustice. The slides warn that **mass surveillance can lead to oppression**.

So, the system would be considered **Halal only under strict conditions**, such as:

- it is used for lawful and beneficial purposes only
- it has strong privacy protections
- it is limited and not used for mass oppression
- there is transparency and oversight
- bias and discrimination are prevented
- data is securely stored and not misused
- it follows justice and public welfare

If the government has a record of targeting minorities or abusing surveillance, then developing such a system becomes highly doubtful or impermissible because preventing harm takes priority over gaining benefits.

In conclusion, AI facial recognition is not automatically Halal or Haram. It can be Halal if it protects life and wealth in a just and limited way, but it becomes impermissible if it violates privacy, supports oppression, or harms the objectives of Shariah.

16. Question:

Apply the method of Qiyas (Analogical Reasoning) to determine whether developing and deploying a Large Language Model (LLM) chatbot that can generate both beneficial and harmful content is permissible in Islam. Clearly identify the four elements of Qiyas — Al-Asl, Al-Far', Al-'Illah, and Al-Hukm — in your analysis, and discuss how the principle of "preventing harm takes precedence over securing benefits" applies.

Answer:

Qiyas (analogical reasoning) is a method used in Islamic law to apply rulings from known cases to new situations. It consists of four elements: **Al-Asl (original case)**, **Al-Far' (new case)**, **Al-'Illah (common cause)**, and **Al-Hukm (ruling)**.

In this case, we analyze whether developing an LLM chatbot is permissible.

- **Al-Asl (Original Case):** Tools like speech, writing, or media platforms, which can be used for both good and bad purposes.
- **Al-Far' (New Case):** LLM chatbot that can generate helpful or harmful content.
- **Al-'Illah (Common Cause):** The ability to influence people and spread information, either beneficial or harmful.
- **Al-Hukm (Ruling):** The ruling depends on usage — permissible if used for good, but prohibited if used for harm.

According to Islamic principles, something that leads to harm can become Haram. The slides also emphasize the rule: **“What leads to Haram is itself Haram”** and **“preventing harm takes precedence over securing benefits.”**

LLM chatbots can provide benefits such as education, assistance, and productivity. However, they can also spread misinformation, generate harmful content, or mislead users. If the harmful effects are not controlled, the system may cause corruption in society.

Therefore, developing such a chatbot is **conditionally permissible (Halal)** only if strict safeguards are implemented, such as:

- content filtering and moderation
- preventing harmful outputs
- transparency about AI usage
- avoiding misuse

If these controls are absent and the system causes widespread harm, then developing or deploying it becomes impermissible.

In conclusion, using Qiyas, LLM chatbots are similar to other dual-use technologies. Their permissibility depends on their purpose and impact, and preventing harm must always come before gaining benefits.

17. Question:

Define Usul al-Fiqh and explain its importance for Muslim software engineers facing new technological challenges. List and explain the four main sources of Islamic law (Quran, Sunnah, Ijma, Qiyas) and illustrate how each can be applied to derive rulings on emerging technologies such as blockchain and cryptocurrency.

Answer:

Usul al-Fiqh means the **principles and methodology used to derive Islamic laws** from primary sources. It helps scholars understand how to apply Islamic teachings to new situations.

For Muslim software engineers, Usul al-Fiqh is very important because modern technologies like blockchain, AI, and cryptocurrency are not directly mentioned in classical texts. Engineers need a structured method to evaluate whether these technologies are Halal or Haram.

The four main sources of Islamic law are:

1. Qur'an

The Qur'an is the primary source of guidance. It provides general principles such as justice, honesty, and prohibition of harm.

Example: In blockchain systems, principles like transparency and fairness can be aligned with Qur'anic values of justice.

2. Sunnah

The Sunnah includes the sayings and actions of the Prophet (PBUH). It explains and clarifies the Qur'an.

Example: Hadith about honesty and trust can guide how financial technologies should be designed ethically.

3. Ijma (Consensus)

Ijma refers to the agreement of qualified scholars on a particular issue.

Example: Scholars may agree on whether certain types of cryptocurrency are permissible based on their characteristics.

4. Qiyas (Analogical Reasoning)

Qiyas applies existing rulings to new cases based on similarity.

Example: Cryptocurrency can be compared to gold or currency. If it meets conditions like real value and no Riba, it may be considered permissible.

Using these sources, Muslim engineers can analyze new technologies carefully. For example, cryptocurrency may be debated because of issues like speculation and uncertainty (Gharar).

In conclusion, Usul al-Fiqh provides a structured framework to make informed decisions. It helps Muslim software engineers apply Islamic principles to modern technologies responsibly and ethically.

18. Question:

Explain the five objectives of Maqasid al-Shariah (Hifz al-Din, Hifz al-Nafs, Hifz al-Aql, Hifz al-Nasl, Hifz al-Mal) with examples from the field of software engineering. How does building or denying quality digital infrastructure relate to these five objectives?

Answer:

Maqasid al-Shariah refers to the higher objectives of Islamic law, which aim to protect five essential aspects of human life: **religion, life, intellect, lineage, and wealth.**

1. Hifz al-Din (Protection of Religion)

This includes supporting religious practices.

Example: Developing Islamic apps such as Qur'an apps or prayer time systems helps people practice their religion.

2. Hifz al-Nafs (Protection of Life)

This focuses on safety and well-being.

Example: Healthcare software, emergency systems, and disaster prediction tools help protect human lives.

3. Hifz al-Aql (Protection of Intellect)

This involves protecting the human mind and knowledge.

Example: Educational platforms and e-learning systems promote learning, while harmful addictive apps may damage intellectual focus.

4. Hifz al-Nasl (Protection of Lineage)

This relates to family structure and dignity.

Example: Avoiding development of inappropriate dating or immoral content platforms helps protect social values and family systems.

5. Hifz al-Mal (Protection of Wealth)

This ensures economic security and fairness.

Example: Secure financial systems, fraud detection, and ethical fintech platforms protect people's wealth.

Building quality digital infrastructure directly supports these objectives. For example:

- Good infrastructure enables access to education (Aql)
- Secure systems protect finances (Mal)
- Healthcare platforms protect life (Nafs)
- Islamic apps support religion (Din)

On the other hand, denying or limiting access to technology can harm society. Poor infrastructure can reduce access to education, healthcare, and economic opportunities.

In conclusion, software engineering plays a major role in achieving the Maqasid al-Shariah. By building ethical and beneficial systems, Muslim engineers can contribute to protecting these five essential objectives of human life.

19. Question:

Discuss the Golden Principle of avoiding doubtful matters in Islam with reference to the Hadith from Sahih al-Bukhari & Muslim. Apply this principle to three gray areas in software development — cryptocurrency platforms, social media algorithms, and dual-use cybersecurity tools — and suggest how a Muslim engineer should navigate these situations.

Answer:

In Islam, one important principle is to avoid doubtful matters. The Hadith says: **“The halal is clear and the haram is clear, and between them are doubtful matters which many people do not know. Whoever avoids doubtful matters protects his religion and honor.”** This Hadith is recorded in **Sahih al-Bukhari and Muslim**.

This principle is very important in modern software development because many technologies are not completely good or completely bad. Some are **gray areas**, meaning they can be used for both benefit and harm. In such cases, a Muslim engineer must be careful, thoughtful, and responsible.

The first gray area is **cryptocurrency platforms**. The slides mention that there is scholarly debate about cryptocurrency. Some concerns include speculation, uncertainty (Gharar), lack of intrinsic value in some coins, and possible use in illegal activities. Therefore, if a platform is designed mainly for risky speculation or doubtful financial practices, a Muslim engineer should avoid

working on it. But if it is designed with transparency, real utility, and without Riba or deception, then it may be more acceptable.

The second gray area is **social media algorithms**. Social media can be used to spread beneficial knowledge, education, and communication, but it can also spread lies, addiction, hatred, and harmful content. A Muslim engineer should examine whether the algorithm promotes truth and benefit or division and corruption. If it is meant to increase engagement by spreading harmful or controversial content, then it should be avoided.

The third gray area is **dual-use cybersecurity tools**. These tools can protect systems and improve security, but they can also be used for hacking and harmful attacks. The slides explain that ethical cybersecurity is allowed, while malicious hacking is forbidden. So, a Muslim engineer should work only on tools that are clearly meant for defense, authorized testing, and protection.

In conclusion, the Golden Principle teaches Muslims to stay away from doubtful matters when the risk of harm is serious. A Muslim software engineer should navigate gray areas by checking intention, impact, possible misuse, and scholarly guidance. If serious doubt remains, the safest choice is to avoid the project.

20. Question:

Using the **Decision-Making Framework** taught in the course, analyze the following scenario: Your company asks you to develop a mass surveillance system for a government that has a record of using technology to oppress minorities. Apply all eight steps of the framework, reference relevant Quranic principles, and state clearly what decision you should make and why.

Answer:

In this case, the correct decision is to **reject the project**, because it is highly likely to support oppression, injustice, and harm. The slides provide a **Decision-Making Framework for Engineers** with eight steps that can be applied here.

Step 1: Is it explicitly mentioned in the Qur'an or Hadith?

There is no direct mention of modern surveillance systems, but the Qur'an clearly forbids injustice, corruption, and harm. The slides also mention that harmful surveillance used for oppression is unethical.

Step 2: What is the purpose behind this work?

The stated purpose may be security, but in this case the government already has a record of oppressing minorities. So the likely real purpose is control and persecution, not justice.

Step 3: Who benefits and who may be harmed?

The government may gain power, but minorities and innocent citizens may suffer serious harm, loss of privacy, fear, and oppression.

Step 4: Does it promote good or evil in society?

This system is more likely to promote injustice, fear, and abuse than social good. The Qur'an says: **"Do not commit abuse on the earth, spreading corruption"** (2:60).

Step 5: Am I being honest and transparent?

If I knowingly build such a system, I become part of a harmful system. That is not honest ethical conduct.

Step 6: Would I be comfortable explaining this to Allah on the Day of Judgment?

No. A Muslim engineer must be able to stand before Allah and answer for what they create. The slides directly ask engineers whether they can stand before Allah for their technology.

Step 7: What do knowledgeable scholars say?

The course teaches that mass surveillance may be permissible only under strict ethical limits, but it becomes dangerous when used for oppression. In this case, the history of abuse makes it clearly unethical.

Step 8: Can I find a Halal alternative?

Yes. Instead of building a mass surveillance system, I could suggest a limited, rights-protecting security system with legal oversight, strict privacy controls,

and protection against abuse. But if the client wants oppression-focused surveillance, I must refuse.

Also, the Islamic legal principles taught in the slides say: “**No harm and no reciprocating harm,**” “**What leads to Haram is itself Haram,**” and “**Preventing harm takes precedence over securing benefits.**” These principles make the decision very clear.

In conclusion, I should **reject the project** because it will likely be used to oppress minorities, violate privacy, and spread injustice. A Muslim engineer must not build technology that supports oppression, even if it is legal or financially rewarding.
